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Methods for developing national biodiversity data for ecosystem accounting in Australia: 1990–2024

A report for the National Ecosystem Accounting Project

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Executive summary

This technical report presents the methods used to develop a timeseries of biodiversity data from 1990 to 2024 for Australia. It provides updated data and methodology to that provided and described by Mokany et al. (2024a, 2024b) and is intended to support national ecosystem accounting in Australia. The method is developed and applied to be consistent with the United Nations System of Environmental Economic Accounting – Ecosystem Accounting (SEEA EA).

Changes since last methodology

The main changes from the previous methodology described by Mokany et al. (2024a, 2024b) are:

- ecosystem condition data input updated to annual coverage (1990–2024) at 100 m spatial resolution
- revised function relating threatened species habitat to ecosystem condition, described in Section 2.2.1, with the underlying supporting research presented in Appendix A
- expanded set of ecosystem accounting areas for which species persistence data were prepared, with the addition of Statistical Area 3 and Local Government Area, as described in Section 2.2.2.

Overview of method

Data were prepared to directly populate ecosystem accounts for terrestrial biodiversity across Australia. All data were developed using a habitat-based approach to biodiversity assessment, which combines spatially complete remotely sensed data on ecosystem condition, with spatially complete information on expected spatial patterns for selected biodiversity features. The data prepared enable reporting for two biodiversity indicators:

- Threatened species habitat: Habitat provision for all nationally listed threatened species with terrestrial distributions.
- Species persistence: The number (or proportion) of originally occurring native species in an ecosystem accounting area expected to persist in the long term anywhere across their original range. Species persistence is reported for four taxonomic groups: vascular plants, reptiles, birds and fungi.

These two indicators represent complimentary biodiversity features and are highly scalable over space and time, enabling consistent and comparable reporting for any set of ecosystem accounting areas. In addition, the utility of both these indicators for accounting for biodiversity within SEEA ecosystem accounts has been demonstrated in peer reviewed scientific publications.

Data sources

The data products were developed using the most credible, suitable, and up-to-date data sources that were available. These data sources were collected using: remotely sensed and on-ground

measurements, modelling techniques and other pre-existing published scientific datasets. The main data sources used to produce the biodiversity data products are described in Section 2.

Limitations and data gaps

The biodiversity data developed here relies on existing spatial datasets as inputs, and is therefore impacted by the accuracy, assumptions and limitations associated with each of those input datasets (Section 2.5). On this basis, the resulting data products are considered the best estimate of expected species persistence and habitat provision for terrestrial threatened species. Given the habitat-based approach applied here, any spatial or temporal inaccuracies in the ecosystem condition data used will directly influence the biodiversity time-series data that have been developed. There are many ways of quantifying biodiversity; here we have focussed on just two indicators, based on the resources available and the suitability of those indicators for providing ecologically meaningful, consistent and rigorous data on biodiversity status and change for the purpose of national ecosystem accounting. Data are only provided for terrestrial biodiversity, given the lack of suitable ecosystem condition spatial time-series data for freshwater aquatic and marine ecosystems.

Data quality

A data quality statement is provided in Section 3. This statement, based on the ABS Data Quality Framework, covers seven dimensions of quality: institutional environment relevance, timeliness, accuracy, coherence, interpretability and accessibility.

1 Introduction

1.1 Background

The *National Ecosystem Accounting Project* (NEAP) is delivering national terrestrial, freshwater and ocean ecosystem accounts, which will help users to understand the importance of the environment and its contribution to our economic and social wellbeing. These accounts will provide a source of trend data for state of the environment reporting and environmental indicators, and may be used to inform the Nature Repair Market. The first release of accounts occurred in February 2025 and was a starting point that will be built on and improved over time as new methods and data become available. The second account release is scheduled for April 2026, and will incorporate updates to methods and data (including this report and associated data collection). NEAP is a collaboration between the Australian Government Department of Climate Change, Energy, the Environment and Water (DCCEEW), the Australian Bureau of Statistics (ABS), and CSIRO.

CSIRO is delivering ecologically meaningful national-scale data, aligned with national and international frameworks, for ecosystem extent and condition, biodiversity and ecosystem services (in physical and monetary terms) for terrestrial, freshwater, coasts and ocean ecosystems. These data will be used as inputs to NEAP to contribute to a set of enduring national accounts aligned with the System for Environmental Economic Accounting – Ecosystem Accounting (SEEA EA). This data can also be used to measure progress against national environmental goals and targets such as the Kunming-Montreal Global Biodiversity Framework.

These data build on the work undertaken in the regional ecosystem account projects, as well as previously released Australian Government environmental economic accounts – including the 2025 National Ecosystem Accounts.

This document is one of a suite of technical methods that describe these data, as well as the process for generating these data (including details on the authoritative datasets used to create these data) and the assumptions and limitations of these data. A full list of technical outputs produced by CSIRO for NEAP is available at https://research.csiro.au/national-ecosystems-and-services-data/.

1.2 About this data collection

This document describes the methods, results, and accuracy, assumptions and limitations for the following data collection in the CSIRO Data Access Portal (DAP):

> Mokany K, Vickers M, Giljohann K, Ware C, Tetreault-Campbell S, Richards AE and Schmidt RK (2025) National biodiversity data for ecosystem accounting in Australia: 1990 to 2024. CSIRO. Data Collection. https://data.csiro.au/collection/csiro:65808

This collection comprises biodiversity data for Australia for use in national ecosystem accounts. The data provided are spatial data and summary tables of predicted status of different biodiversity features across Australia over time. Data are provided separately for two types of biodiversity

features: (1) nationally listed threatened species; (2) all native species in broad taxonomic groups (vascular plants, birds, reptiles, fungi). The status of all biodiversity features has been estimated annually, for each year from 1990 to 2024.

The data are generated using a habitat-based biodiversity assessment approach, where expected spatial patterns of biodiversity in reference condition are combined with a time series of spatial patterns in ecosystem condition. For all biodiversity features, annual patterns of expected (or potential) status were derived by combining remotely sensed data with information on inferred reference patterns in biodiversity (pre-European), incorporating field observation data.

These data align with other data prepared for Australia under the *National Ecosystem Accounting Project*, including for ecosystem extent, ecosystem condition and ecosystem services.

The biodiversity data comprise spatial raster layers, shapefiles and summary tables of predicted status of different biodiversity features across Australia over time. Also provided are supplementary data tables, maps and charts.

1.3 About this document

This document contains:

- methods for developing data
  - concepts and conceptual models (Section 2.1)
  - the data processing steps applied for developing data (Section 2.2)
- the results from applying the methods
  - a list of the figures and data products that are part of this collection (sections 2.3, 2.4)
- accuracy, assumptions and limitations (Section 2.5).
- data quality statement (Section 3).

2 Method

2.1 Concepts and conceptual models

2.1.1 Spatial data

The SEEA ecosystem accounting framework (SEEA EA) defines several technical concepts for ecosystem accounting, which relies on data that may be spatially and/or temporally dynamic.

- **Basic Spatial Units (BSUs)** are the smallest unit of aggregation that are used consistently across accounts and against which stocks and flows are recorded. The BSUs are often represented by a regular grid (e.g. raster images) and are used in the development of 'account-ready' data. These are data that have been prepared such that they can be used (nearly) directly (with minimal additional processing) for compiling ecosystem accounts.
- The **Ecosystem Accounting Area (EAA)** defines the spatial boundary for accounting. The EAA ensures that the various stocks and flows are assessed consistently and must not change for any set of coherent accounts.

Each **BSU** must contain mutually exclusive attributes (e.g. ecosystem types, carbon stocks) and provide exhaustive coverage (i.e. tessellation) of the **EAA**.[1] This mutually exclusive nature ensures that each attribute is uniquely assigned to one BSU, preventing overlap and redundancy.

Further detail is provided in the methods for developing BSUs (Liu et al. 2025).

2.1.2 Biodiversity

Biodiversity is 'the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems' (Convention on Biological Diversity (CBD), article 2, entitled "Use of Terms"). Within the Commonwealth's *Environment Protection and Biodiversity Conservation Act 1999* the 'components of biodiversity includes species, habitats, ecological communities, genes, ecosystems and ecological processes' (Ch 5, Part 12, Div. 1).

Accounting for biodiversity is presented as thematic accounting in the SEEA EA framework. In contrast to accounting for ecosystem extent, ecosystem condition and ecosystem services, the recommendations on accounting for biodiversity have not been ratified as a standard in the SEEA EA framework. Some guidance provided in the SEEA EA in Chapter 13 (United Nations et al. 2024) show how biodiversity is related to the parts of ecosystem accounting that have been standardised. Guidance has also been published on addressing spatial scale in deriving biodiversity accounts under SEEA Ecosystem Accounting (Larsen et al. 2021). For Australia, biodiversity accounts have been prepared previously for focal regions, as part of the *Regional Ecosystem Accounting Pilot* project (Giljohann et al. 2023a, 2023b; Mokany et al. 2023; Schmidt et al. 2023b,

[1] Contiguous BSUs of the same ecosystem type may be combined when accounting for individual Ecosystem Assets (EAs), such as individual wetlands of forest stands, is of concern. EAs are also a spatial statistical unit described in the SEEA EA.

2023a; Schmidt et al. 2023c). Biodiversity data has been prepared previously for national ecosystem accounts (Mokany et al. 2024b) though these data were not used by the Australian Bureau of Statistics in their publication *National Ecosystem Accounts - experimental estimates* (ABS 2025).

For the present publication, data were prepared with the objective of directly populating ecosystem accounts for terrestrial biodiversity across Australia. All data were developed using a habitat-based approach to biodiversity assessment (King et al. 2016; King et al. 2021; Mokany et al. 2022d; Mokany et al. 2024b). This habitat-based approach combines spatially complete remotely sensed data on ecosystem condition developed by the Habitat Condition Assessment System (HCAS)[2] (Valavi et al. 2025a; Williams et al. 2025), with spatially complete information on expected spatial patterns for selected biodiversity features. Data are provided that enable reporting for two biodiversity indicators (Table 1):

- **Threatened species habitat**: Habitat provision for all nationally listed threatened species with terrestrial distributions, expressed as either a quantity (species hectares) or as a proportion of the estimated historic (pre-European; 1750) amount (Giljohann et al. 2025).
- **Species persistence**: The number (or proportion) of originally occurring native species in the ecosystem accounting area expected to have been retained and to persist in the long term anywhere across their original range, assuming no further changes in ecosystem condition. This indicator is expressed as either a quantity (number of species) or as a proportion of the estimated historic (pre-European; 1750) amount. This indicator is equivalent to the Biodiversity Habitat Index (UNEP-WCMC 2023), quantifying the consequences of the level of habitat provision in a given year on the expected ongoing persistence of species into the future.

For the species persistence indicator, data are provided for four taxonomic groups: vascular plants, reptiles, birds and fungi. For each of these taxonomic groups, the assessment is derived from community-level biodiversity models that consider all Australian native species in that taxonomic group (described below).

These two indicators were selected as they represent complimentary biodiversity features. Both indicators are highly scalable over space and time, enabling consistent and comparable reporting for any set of ecosystem accounting areas. The indicators also link directly to ecosystem condition data (Valavi et al. 2025a; Williams et al. 2025), enabling an integrated approach and the potential for exploratory analyses of change over space and time. In addition, both indicators have demonstrated utility in accounting for biodiversity within SEEA ecosystem accounts (Giljohann et al. 2025; King et al. 2021; Mokany et al. 2022d; Mokany et al. 2024b; Schmidt et al. 2023b, 2023a).

[2] https://research.csiro.au/biodiversity-knowledge/projects/hcas/

Table 1 Summary of the biodiversity features and metrics available in the data publication

INDICATOR	METRIC NAME	METRIC DESCRIPTION	UNITS	BIODIVERSITY FEATURES
Threatened species habitat	Threatened species habitat	Habitat provision for nationally listed threatened species	species.hectares	Threatened species
	Proportion of threatened species habitat remaining	Proportion of pre-European (1750) habitat provision for nationally listed threatened species	NA	
Species persistence	Number of species expected to persist	The number of originally occurring species in the ecosystem accounting area expected to persist in the long term anywhere across their original range	species	Vascular plant species Reptile species Bird species Fungi species
	Proportion of species expected to persist	The proportion of originally occurring species in the ecosystem accounting area expected to persist in the long term anywhere across their original range	NA	

NA = not applicable – these are proportions which scale from 0 to 1, without units

2.2 Data processing

2.2.1 Threatened species habitat

The data for the terrestrial distributions of all nationally listed threatened species (*Environment Protection and Biodiversity Conservation Act 1999*) were derived by combining two key data sources (Figure 2):

- estimated pre-European (1750) spatial distributions for all threatened species (Giljohann et al. (2022) - Release 3, 2025)
- annual spatial layers for the time series of ecosystem condition across Australia (100 m resolution) derived from the HCAS version 3.3 data collection (90 m resolution) - ecosystem site condition (Valavi et al. 2025a; Williams et al. 2025).

Estimated pre-European (1750) terrestrial spatial distributions for 1,785 nationally listed threatened species were derived based on a combination of data sources (Giljohann et al. (2022) - Release 3, 2025) , including occurrence observations for all years from the Atlas of Living Australia (Atlas of Living Australia 2025) and mapped distributions from the Australian Government's Species of National Environmental Significance Database (DCCEEW 2025a) (Figure 2). Using refined distribution information (removing outliers; Giljohann et al. (2022)), each species' estimated pre-European distribution was derived using a modified version of the Area of Habitat protocol (Brooks et al. 2019), which involved intersecting occurrence data with fine resolution (100 m) spatial layers of pre-1750 vegetation mapping (DCCEEW 2021), IBRA bioclimatic subregions (DCCEEW 2020) and elevation (Hutchinson et al. 2008). Spatial limits were then applied to the areas of identified habitat (elevation bands within vegetation types within IBRA sub-regions) to derive the final maps of estimated pre-European distributions for each species (Giljohann et al. 2025; Giljohann et al. 2022). The estimated pre-European distribution maps for all 1,785 species were combined to produce a single spatial layer for the number of threatened species with estimated pre-European distributions in each terrestrial grid cell (100 m resolution) (Figure 2).

To derive the threatened species habitat data for each year, both the ecosystem site condition raster data (Valavi et al. 2025a; Williams et al. 2025) and the number of threatened species raster data ('Release_3_2025/realm_summaries/grids/terrestrial_historical_sum.tif' from Giljohann et al. (2022)) were reprojected to the 100 m resolution BSU raster (Liu et al. 2025) using a weighted average of all cells in the source raster that overlap with a single 100 m BSU raster cell. The ecosystem site condition (hereafter 'ecosystem condition') represents an indicator of ecosystem condition for each grid cell, with values ranging continuously from 0 (fully degraded) to 1 (reference condition). The single spatial raster layer of the number of species with pre-European distributions overlapping each 100 m grid cell ('ThreatenedSpeciesHabitat_1750.tif') was then multiplied by the 100 m resolution annual ecosystem condition raster data for each year ('EcosystemConditionInput_<YYYY>.tif' for YYYY in 1990 to 2024) to derive the estimated threatened species habitat raster data for each year ('ThreatenedSpeciesHabitat_<YYYY>.tif').

This approach to deriving the threatened species habitat data assumes that the value of an area in providing habitat for threatened species is on average directly proportional to the estimated ecosystem condition of that area (Figure 1). The validity of this assumption was explored through two avenues: (i) a literature review, and; (ii) a quantitative assessment using threatened species population data from the threatened species index (TSX 2025) combined with remotely derived ecosystem condition data (Valavi et al. 2025b) (Appendix A). In summary, very few published studies were found to have investigated the relationship between species population size and ecosystem condition in Australia (Appendix A). The quantitative assessment of threatened species index data undertaken here was limited by the small number of species for which suitable data were available (Appendix A). However, there was clear evidence of some threatened species persisting in locations with very low estimated ecosystem condition (Appendix A). Given this review and assessment, the simplest and most appropriate approach was determined to be to assume a direct linear (1:1) relationship between ecosystem condition of an area and its value (on average) as habitat for threatened species (Figure 1).

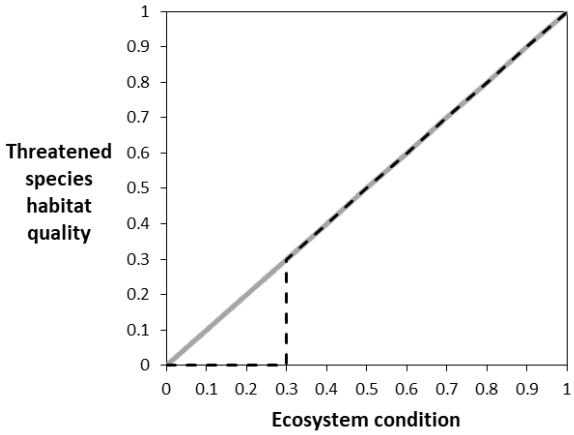

Figure 1 The relationship between the quality of habitat for supporting threatened species and ecosystem condition
Previous applications of the threatened species habitat indicator (Giljohann et al. 2025; Mokany et al. 2024b) have applied a threshold function, where locations with ecosystem condition below a threshold (e.g. 0.3) were assumed to provide no value in supporting threatened species (dashed black line). The alternative approach which has been adopted and applied for the data derived here assumes a linear 1:1 relation (solid grey line) given the evidence presented in Appendix A.

The resulting spatial layer (ThreatenedSpeciesHectares_<YYYY>.tif) provides the estimated effective 'species hectares' of habitat being provided to nationally listed threatened species for

each year. Grid cells with lower ecosystem condition provide less effective habitat area for threatened species (species hectares, defined as the number of threatened species multiplied by the effective habitat area) given the same number of threatened species with pre-European distributions in that location.

Note that an alternative derivation of the threatened species habitat indicator could be based on estimated extant (current) areas of habitat potentially utilised by each threatened species (Giljohann et al. (2022) - Release 3, 2025), rather than the historic distributions as used here. Such an approach would give greater emphasis to the areas currently most important in supporting threatened species. However, because such an approach when used in ecosystem accounting would overlook past loss of threatened species habitat, here we have used estimated historic distributions.

In addition, it should be noted that the threatened species habitat data product reflects habitat for the species listed as threatened on 28th May 2025 (Giljohann et al. (2022) - Release 3, 2025). Ongoing changes in the species listed as threatened will require future versions of this data product to derive the full time series based on updated versions of the threatened species estimated historic distributions (Giljohann et al. (2022) - Release 3, 2025).

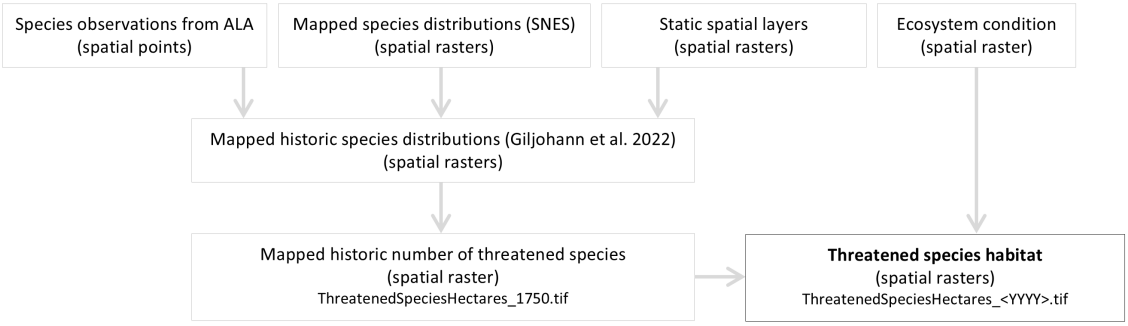

Figure 2 Conceptual depiction of the input data derivation process for nationally listed terrestrial threatened species

In using the threatened species habitat data to populate ecosystem accounts, it is most appropriate to sum the values of the amount of threatened species habitat in each grid cell within an ecosystem accounting area. This summary approach retains the original units (species.hectares).

2.2.2 Species persistence

The compiled biodiversity account data for the expected species persistence within four taxonomic groups were derived using the community-level assessment method as described in Mokany et al. (2022d), used in a range of previous studies (Allnutt et al. 2008; Di Marco et al. 2019; Ferrier et al. 2004; UNEP-WCMC 2016), and adopted as a component indicator (the Biodiversity Habitat Index) under the Convention on Biological Diversity Kunming-Montreal Global Biodiversity Framework (CBD 2022). This indicator has a range of benefits, including supporting accounting for the non-additive nature of biodiversity – that the persistence of species within an area often depends partially on habitat availability outside that area (Larsen et al. 2021). The

approach is also beneficial in considering large numbers of species simultaneously; effectively all the species within the taxonomic group being assessed. As species persistence is an emergent property of the habitat available by BSUs in an area and its surrounds, it is estimated at the ecosystem accounting area scale.

The analytical approach to estimate the proportion of species in the ecosystem accounting area that are expected to persist over the long term (assuming spatial estimates of ecosystem condition at a time point remain unchanged for all years thereafter), has 4 main inputs:

- a spatial layer of ecosystem condition at a time point (250 m resolution) derived from the HCAS version 3.3 data collection (90 m resolution) - ecosystem site condition (Valavi et al. 2025a; Williams et al. 2025) (Figure 3(a))
- spatial layers from a generalized dissimilarity model (GDM) (Ferrier et al. 2007) (Figure 3(h)) enabling the prediction of compositional similarity under reference ecosystem condition between any 2 locations (Mokany 2022; Mokany et al. 2022c)
- a spatial layer of expected species richness under reference ecosystem condition (Figure 3(i)) (Mokany 2022; Mokany et al. 2022c)
- a spatially defined ecosystem accounting area (Figure 3(d)) (Section 2.2.2).

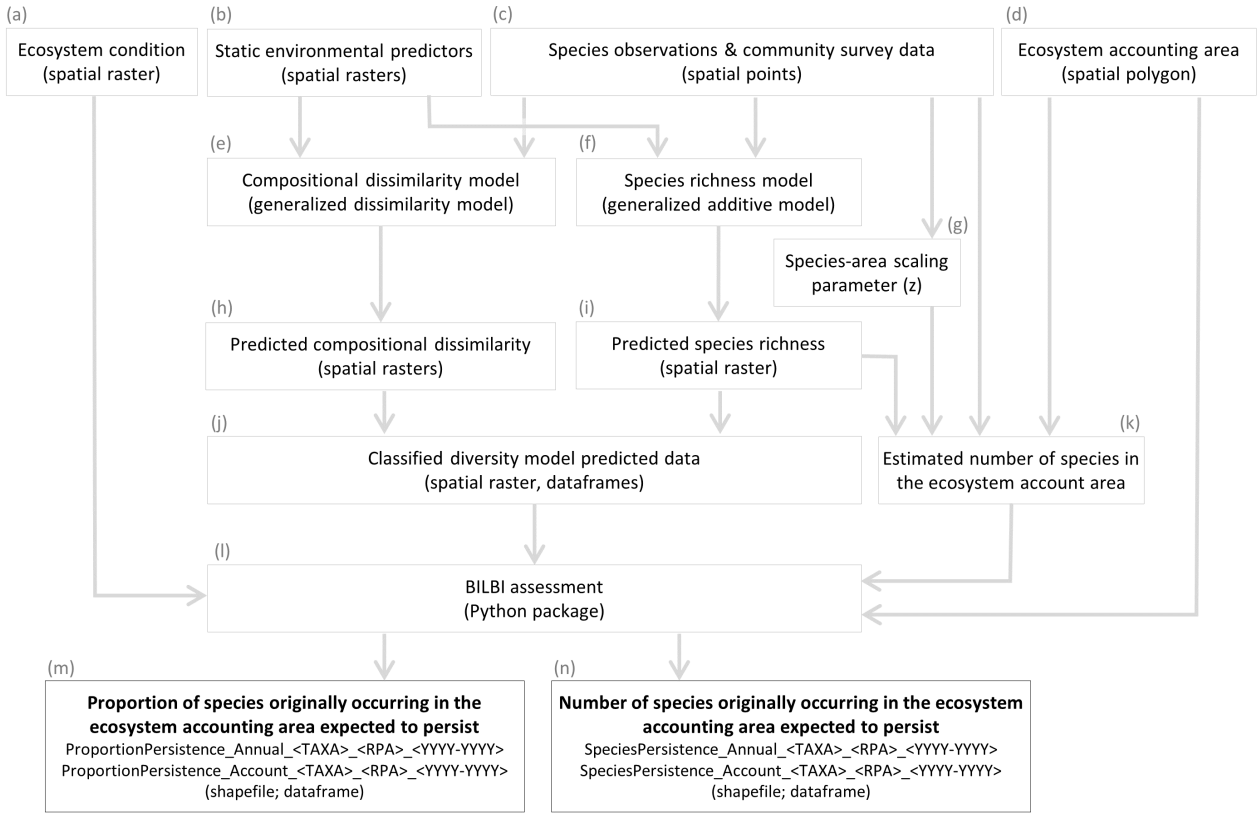

Figure 3 Conceptual depiction of the data processing associated with the derivation of the expected species persistence metrics ((m) and (n))

Given all the biodiversity model predicted spatial layers were available at 250 m grid resolution, the ecosystem condition data were resampled from their native resolution (90 m) (Valavi et al. 2025a; Williams et al. 2025) to 250 m resolution, based on a weighted average (where the weight is the proportion of each 90 m grid cell within each 250 m grid cell). All species persistence calculations described were therefore undertaken using the aligned 250 m input spatial layers.

For each taxonomic group considered, this approach uses spatial biodiversity models for two aspects of diversity: species richness (Figure 3 (f)) and pairwise compositional similarity (Figure 3(e)). For computational efficiency, the spatial compositional similarity models are classified into 10,000 classes by k-means clustering (Ikotun et al. 2023) (Figure 3(j)) for which the grid cells within a class are predicted to be highly compositionally similar. The classified summaries of the diversity models are then used as inputs by the analysis (in CSIRO's 'nature-positive' Python package (Figure 3(l))), along with the grided ecosystem condition raster (Figure 3(a)), and the spatial representation of the ecosystem accounting area (polygon) (Figure 3(d)) .

The primary output from the species persistence assessment (Figure 3(l)) is the proportion of species originally occurring in the ecosystem accounting area (pre-European) that are expected to persist indefinitely into the future anywhere in their range given the spatial patterns in ecosystem condition in a given year (Figure 3(m)). This proportion value is converted into an estimate of the number of species expected to persist (Figure 3(n)), by multiplying the proportion of species persisting by the estimated number of native species expected to have originally occurred in the ecosystem accounting area (Figure 3(k)). For most taxa and ecosystem accounting areas, this estimated number of species originally occurring is calculated as the maximum of two alternative methods: (i) the number of native species in the specified taxonomic group with recorded occurrences in the ecosystem accounting area (Atlas of Living Australia 2024); (ii) the estimated number of species from the species-area relationship (power model) (Preston 1962), applying mean local richness from the species richness prediction raster and power-model scaling factor (z) derived from ecological survey data. For state/territory and national ecosystem accounting areas, the estimates of the number of originally occurring species were constrained to be no less than reliable published catalogues of the number of native species in those jurisdictions (ABRS 2024; Australian Plant Census 2024).

The species persistence assessment provides estimates of the total number or proportion of species expected to persist for each year of ecosystem condition data. The assessment also provides a nuanced assessment of the nature of the change in estimated species persistence from one year to the next. As per the SEEA Ecosystem Accounting framework (United Nations et al. 2024) this includes a breakdown of the total change due to additions and reductions, which are determined by separately considering the change in expected species persistence due to grid cells (BSUs) which have increased in ecosystem condition, and those which have decreased in ecosystem condition. The species persistence assessment provides a further breakdown of the change in species persistence, providing information on the amount due to changes in ecosystem condition within the ecosystem accounting area as well as the changes in species persistence due to changes in ecosystem condition outside the ecosystem accounting area (Table 2).

Due to continuous spatial change in the distributions of species and the composition of communities, the expected persistence of species occurring within an ecosystem accounting area cannot be determined without considering changes in ecosystem condition outside the ecosystem accounting area. The resultant account table for species persistence (Table 2) is created by a separate analysis for each change component, including application of a power scalar (z) aligned with the power model of the species-area relationship ($S = cA^z$). Because of this important non-linear transformation, the component values within the account for species persistence are often not additive. This non-additive aspect of the species persistence account aligns with well established ecological theory and knowledge (Drielsma et al. 2014; Ferrier and Drielsma 2010;

Larsen et al. 2021), including the fact that species will often occur across multiple ecosystem accounting areas, though requires care in interpretation and use of the data. The expected species persistence values for an ecosystem accounting area cannot be derived by summing the persistence values for multiple smaller ecosystem accounting areas that sit within it, but instead requires a new species persistence analysis.

Table 2 Definition of the accounting components within the species persistence account data for the four taxonomic groups considered

Account component	Definition	Shapefile field name
pre-European (1750)	The estimated number of native species originally occurring in the ecosystem accounting area.	Open1750
Opening value	The estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given spatial patterns in ecosystem condition in the opening year.	OpenYYYY (e.g. Open1990)
Additions due to increases in condition within the ecosystem accounting area	The change in the estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given all increases in ecosystem condition within the ecosystem accounting area from the opening year to the closing year.	AddWitYYYY (e.g. AddWit1991)
Additions due to increases in condition outside the ecosystem accounting area	The change in the estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given all increases in ecosystem condition outside the ecosystem accounting area from the opening year to the closing year.	AddOutYYYY (e.g. AddOut1991)
Total additions	The change in the estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given all increases in ecosystem condition from the opening year to the closing year.	AddTOTYYYY (e.g. AddTOT1991)
Reductions due to decreases in condition within the ecosystem accounting area	The change in the estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given all decreases in ecosystem condition within the ecosystem accounting area from the opening year to the closing year.	RedWitYYYY (e.g. RedWit1991)
Reductions due to decreases in condition outside the ecosystem accounting area	The change in the estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given all decreases in ecosystem condition outside the ecosystem accounting area from the opening year to the closing year.	RedOutYYYY (e.g. RedOut1991)
Total reductions	The change in the estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given all decreases in ecosystem condition from the opening year to the closing year.	RedTOTYYYY (e.g. RedTOT1991)
Net Change due to changes in condition within the ecosystem accounting area	The change in the estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given all changes in ecosystem condition within the ecosystem accounting area from the opening year to the closing year.	ChWitYYYY (e.g. ChWit1991)
Net Change due to changes in condition outside the ecosystem accounting area	The change in the estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given all changes in ecosystem condition outside the ecosystem accounting area from the opening year to the closing year.	ChOutYYYY (e.g. ChOut1991)
Total net change	The change in the estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given all change in spatial patterns in ecosystem condition from the opening year to the closing year.	ChTOTYYYY (e.g. ChTOT1991)
Closing value	The estimated number of native species in the ecosystem accounting area that are expected to persist in the long-term, anywhere across their range, given spatial patterns in ecosystem condition in the closing year.	CloseYYYY (e.g. Close1991)

The data inputs for the assessment of species persistence for each taxonomic group considered (vascular plants, reptiles, birds and fungi) differ, due to the nature of the available source data. The inputs for each taxonomic group are described in the following sections.

Vascular plants

For vascular plants, spatial biodiversity models of species richness and pairwise compositional similarity were derived using plant community composition data from 115,086 survey plots across Australia (Mokany et al. 2022c; Mokany et al. 2022b). Plant species richness was modelled across Australia using generalized additive modelling (Hastie and Tibshirani 1986) with nine spatial environment layers as predictors (Mokany et al. 2022b). Plant community compositional similarity was modelled using generalized dissimilarity modelling (Ferrier et al. 2007) with eight spatial environment layers as predictors, plus geographic distance (Mokany et al. 2022b).

To enable estimation of the number of native species originally occurring (pre-European) in each ecosystem accounting area, all available species observation records were obtained from the Atlas of Living Australia on 11 June 2024 (Atlas of Living Australia 2024), filtered by year (≥1970) and spatial accuracy (≤2,000m), and aggregated to 250 m grid resolution. The resulting data covered 32,761,945 gridded species observations for 19,875 native Australian plant species. The species-area relationship power model scalar (z in: $S = cA^z$) for vascular plants in Australia was estimated using all available data (vegetation surveys and the gridded occurrence data). For a range of areas around each survey location (1 ha, 100 ha, 500 ha, 1000 ha, 5000 ha, 10,000 ha, 25,000 ha, 50,000 ha, 75,000 ha, 100,000 ha) the total number of native vascular plant species observed was determined, then the species-area scalar (z) estimated by non-linear least squares (nls) fitting in R (R Core Team 2024), with an estimated value of z = 0.219.

Reptiles

For reptiles, spatial biodiversity models of species richness and pairwise compositional similarity were derived by Mokany (2022) using 144,196 occurrence observations of 888 Australian native reptile species (Atlas of Living Australia 2022) filtered by year (≥1970) and spatial accuracy (≤2,000 m) and aggregated within 18,556 grid cells (~250 m resolution). The 18,556 locations with sufficient biological records to constitute a community sample were selected using a spatial filtering approach described in Mokany et al. (2022). Reptile species richness was modelled across Australia using generalized additive modelling with nine spatial environment layers as predictors (Mokany 2022). Reptile community compositional similarity was modelled using generalized dissimilarity modelling with nine spatial environment layers as predictors, plus geographic distance (Mokany 2022).

To enable estimation of the number of native species originally occurring (pre-European) in each ecosystem accounting area, an updated set of all available native reptile species observation records were obtained from the Atlas of Living Australia on 4 June 2024 (Atlas of Living Australia 2024), filtered by year (≥1970) and spatial accuracy (≤2,000 m), and aggregated to 250 m grid resolution. The resulting data covered 567,962 gridded species observations for 927 native Australian reptile species. The species-area relationship power model scalar (z in: $S = cA^z$) for reptiles in Australia was estimated using the gridded species observation data using the same method as for vascular plants, producing an estimated species-area scalar value of z = 0.187 for reptiles in Australia.

Birds

For birds, spatial biodiversity models of species richness and pairwise compositional similarity were derived by Mokany (2022) using 2,597,530 occurrence observations of 749 Australian native bird species (Atlas of Living Australia 2022) filtered by year (≥1970) and spatial accuracy (≤2,000m), and aggregated within 64,002 grid cells (~250 m resolution). The 64,002 locations with sufficient biological records to constitute a community sample were selected using a spatial filtering approach described in Mokany et al. (2022). Bird species richness was modelled across Australia using generalized additive modelling with nine spatial environment layers as predictors (Mokany 2022). Bird community compositional similarity was modelled using generalized dissimilarity modelling with nine spatial environment layers as predictors, plus geographic distance (Mokany 2022).

To enable estimation of the number of native species originally occurring (pre-European) in each ecosystem accounting area, an updated set of all available native bird species observation records were obtained from the Atlas of Living Australia on 12 June 2024 (Atlas of Living Australia 2024), filtered by year (≥1970) and spatial accuracy (≤2,000 m), and aggregated to 250 m grid resolution. The resulting data covered 7,393,708 gridded species observations for 738 native Australian bird species. The species-area relationship power model scalar (z in: $S = cA^z$) for birds in Australia was estimated using the gridded species observation data using the same method as for vascular plants, producing an estimated species-area scalar value of z = 0.169 for birds in Australia.

Fungi

For fungi, spatial biodiversity models of species richness and pairwise compositional similarity were derived by Mokany (2022) using genomically analysed soil data from 385 sites across Australia, collected by the Biodiversity of Australian Soils (BASE) project (Bissett et al. 2016). The compositional data are Operational Taxonomic Unit (OTU) assignments, which were assumed to be equivalent to species. Fungi species richness was modelled across Australia using generalized additive modelling with four spatial environment layers as predictors (Mokany 2022). Fungi community compositional similarity was modelled using generalized dissimilarity modelling with eight spatial environment layers as predictors, plus geographic distance (Mokany 2022).

To enable estimation of the number of native species originally occurring (pre-European) in each ecosystem accounting area, a set of all available fungi species observation records were obtained from the Atlas of Living Australia on 13 June 2024 (Atlas of Living Australia 2024), filtered by year (≥1970) and spatial accuracy (≤2,000 m), and aggregated to 250 m grid resolution. The resulting data covered 146,579 gridded species observations for 2,234 native Australian fungi species. The species-area relationship power model scalar (z in: $S = cA^z$) for fungi in Australia was estimated using the same method as for vascular plants, producing an estimated species-area scalar value of z = 0.249 for fungi in Australia.

Ecosystem accounting areas

Account data for species persistence were prepared for a number of different sets of ecosystem accounting areas, all within the bounds of Australia's contiguous continental Exclusive Economic Zone (EEZ) (Liu et al. 2025). These ecosystem accounting areas include:

- **National** – all areas within Australia's contiguous continental EEZ (Liu et al. 2025). Reporting is for all areas together. Data file label for ecosystem accounting area (<EAA>) = 'National'.
- **Australian States and Territories** – each of the 9 Australian States and Territories, including 'other' (Australian Bureau of Statistics (2021) ; field = STE_NAME21). Data file label for ecosystem accounting area (<EAA>) = 'StateTerritory'.
- **Statistical Area 2** – each of the 2,451 Australian Statistical Area Level 2 (SA2) regions (Australian Bureau of Statistics (2021) ; field = SA2_NAME21). Data file label for ecosystem accounting area (<EAA>) = 'SA2'. By request, results for SA2 have also been provided using the code for each SA2 region (field = SA2_CODE21). Data file label for ecosystem accounting area (<EAA>) = 'SA2code'.
- **Statistical Area 3** – each of the 336 Australian Statistical Area Level 3 (SA3) regions (Australian Bureau of Statistics (2021) ; field = SA_NAME21). Data file label for ecosystem accounting area (<EAA>) = 'SA3'.
- **Local Government Areas** – each of the 546 Australian Local Government Areas (LGA) (Australian Bureau of Statistics (2021) ; field = LGA_NAME25). Data file label for ecosystem accounting area (<EAA>) = 'LGA'.
- **IBRA regions** – each of the 86 regions of the Interim Biogeographic Regionalisation for Australia (IBRA) v7.1 (DCCEEW (2025b); field = REG_NAME_7) that cover the continent of Australia and nearby islands. This excludes two IBRA regions that cover offshore islands, for which biodiversity and ecosystem condition data were unavailable ("Coral Sea", "Pacific Subtropical Islands"). Data file label for ecosystem accounting area (<EAA>) = 'IBRA'.
- **IBRA sub-regions** – each of the 412 subregions of the Interim Biogeographic Regionalisation for Australia (IBRA) v7.1 (DCCEEW (2025b); field = SUB_NAME_7) that cover the continent of Australia and nearby islands. This excludes two IBRA subregions that cover offshore islands, for which biodiversity and ecosystem condition data were unavailable ("Coral Sea", "Lord Howe Island"). Data file label for ecosystem accounting area (<EAA>) = 'IBRAsub'.
- **NRM regions** – each of the 57 Natural Resource Management (NRM) regions (DCCEEW (2023); field = NRM_REGION) that cover the continent of Australia and nearby islands. This excludes 3 NRM regions that cover offshore islands or marine areas, for which biodiversity and ecosystem condition data were unavailable ("Marine NRM"; "Ashmore and Cartier Islands"; "North Coast - Lord Howe Island"). Data file label for ecosystem accounting area (<EAA>) = 'NRM'.
- **Drainage divisions** – each of the 13 Australian Water Resources Assessment - Geofabric Drainage Divisions from the Australian Hydrological Geospatial Fabric (Geofabric) (Bureau of Meteorology (2023) ; field = AWRADrainageDivision). Data file label for ecosystem accounting area (<EAA>) = 'DrainageDivision'.
- **River regions** – each of the 219 river region catchment areas ("RiverRegion") from the Geofabric (Bureau of Meteorology (2023) ; field = RiverRegion). Data file label for ecosystem accounting area (<EAA>) = 'RiverRegion'.
- **Protected areas (all)** – all areas together that are in any protected area category in the Collaborative Australian Protected Areas Database (CAPAD) 2024 - Terrestrial ((DCCEEW 2024b); field = IUCN). Reporting is for all protected areas in combination. Data file label for ecosystem accounting area (<EAA>) = 'ProtectedAreasAll'.

- **Protected areas (IUCN)** – each of the 8 IUCN categories of protected area, including the three non-IUCN categories (NR, NA, NAS) combined, in the Collaborative Australian Protected Areas Database (CAPAD) 2024 - Terrestrial ((DCCEEW 2024b); field = IUCN). Data file label for ecosystem accounting area (<EAA>) = 'ProtectedAreasIUCN'.
- **50 km gridded** – each of the 3,399 gridded areas of 50 x 50 km across the terrestrial areas of Australia. Data file label for ecosystem accounting area (<EAA>) = 'Gridded50km'.

All feature data were projected into GDA94 geographic projection (EPSG:4283) prior to analysis, then outputs projected to GDA2020 Albers (EPSG:9473) for alignment with the BSU projection (Liu et al. 2025).

2.3 Summary figures

The biodiversity data were summarised as image format for basic viewing of spatial and temporal trends in the data (Table 4). These summary images include:

- **Threatened species habitat**
  - chart of total threatened species habitat amount across all Australia over the time series (1990–2024)
  - chart of proportion of historic (pre-European: 1750) threatened species habitat remaining across all Australia over the time-series (1990–2024)
  - annual map images of proportion of historic (pre-European: 1750) threatened species habitat remaining (1990–2024)
  - map images of change in threatened species habitat amount between adjacent years (1990–2024)
  - map images of change in proportion of historic (pre-European: 1750) threatened species habitat remaining between adjacent years (1990–2024).
- **Species persistence**
  - chart of the number of species expected to persist across all Australia over the time series (1990–2024), for each taxonomic group
  - chart of the proportion of originally occurring species expected to persist across all Australia over the time series (1990–2024), for each taxonomic group
  - annual map images of the number of species expected to persist for each taxonomic group (1990–2024) within: (i) each Statistical Area 2; (ii) each 50 x 50 km area.
  - annual map images of the proportion of species expected to persist for each taxonomic group (1990–2024) within: (i) each Statistical Area 2; (ii) each 50 x 50 km area.
  - annual map images of change in the number of species expected to persist for each taxonomic group between adjacent years (1990–2024), within: (i) each Statistical Area 2; (ii) each 50 x 50 km area.
  - annual map images of change in the proportion of species expected to persist for each taxonomic group between adjacent years (1990–2024), within: (i) each Statistical Area 2; (ii) each 50 x 50 km area.

2.4 Data dictionary and storage structure

The data products provided as part of this collection are described in the following sections. These were generated by applying the methods described in Section 2.2.

2.4.1 Data dictionary

The data products in this collection are described in Table 3. These were generated by applying the methods described in Section 2.2 and, unless indicated, meet the following specifications:

Raster data

- Spatial reference: GDA2020 / Australian Albers (EPSG:9473)
- Spatial resolution: 100 m
- Raster file format: GeoTiff
- Spatial extent: xmin: -2253900, xmax: 2949600, ymin: -5218400, ymax: -929100

Feature data

- Coordinate reference system: GDA2020 / Australian Albers (EPSG:9473)
- File format: shapefile and GeoPackage

Tabular data

- File format: comma-separated value (csv)

2.4.2 Image file dictionary

The data products in this collection were summarised as images described in Table 4. These were generated by applying simple mapping and plotting functions in R (R Core Team 2025).

2.4.3 Storage structure

Data in this collection are organised in the following folder structure.

- 🗀 ThreatenedSpeciesHabitat
  - 🗀 threatened_species_habitat_data

    [annual rasters (1750; 1990-2024) of threatened species habitat]
  - 🗀 ecosystem_condition_input

    [annual rasters (1990-2024) of input ecosystem condition (100 m resolution) used for the threatened species habitat calculation]
  - 🗀 images

    [map images and time-series charts of threatened species habitat]
- 🗀 Persistence
  - 🗀 <EAA> [each of: 'National'; 'StateTerritory'; 'SA2'; 'SA2code'; 'SA3', 'LGA', 'IBRA'; 'IBRAsub'; 'NRM'; 'DrainageDivision' ; 'RiverRegion' ; 'ProtectedAreasAll' ; 'ProtectedAreasIUCN' ; 'Gridded50km']
    - 🗀 <TAXA> [each of: 'Birds'; 'Fungi'; 'Plants'; 'Reptiles']
      - 🗀 SpeciesPersistence
        - 🗀 Annual
        - 🗀 Account
      - 🗀 ProportionPersistence
        - 🗀 Annual
        - 🗀 Account
  - 🗀 ecosystem_condition_input

    [annual rasters (1990-2024) of input ecosystem condition (250 m resolution) used for the species persistence calculations]
  - 🗀 images

    [map images and time-series charts of species persistence]
    - 🗀 <EAA> [each of: 'National'; 'SA2'; 'Gridded50km']

Table 3 Data dictionary for biodiversity data

METRIC NAME	DESCRIPTION	SUMMARY TYPE	UNITS	DATA TYPE	DATA FORMAT	FILE NAME TEMPLATE (WITH EXAMPLE)
Threatened species habitat	Habitat provision for nationally-listed threatened species	annual	species.hectares	raster	GeoTIFF (.tif)	ThreatenedSpeciesHabitat_<YYYY>.tif (e.g. ThreatenedSpeciesHabitat_2024.tif)
Number of species expected to persist	The number of originally occurring species in the reporting region expected to persist in the long term anywhere across their original range assuming no further change in ecosystem condition	total for all years	species	data frame	comma-separated value (.csv)	SpeciesPersistence_Annual_<TAXA>_<EAA>_<YYYY-YYYY>.csv (e.g. SpeciesPersistence_Annual_Plants_IBRA_1990-2024.csv)
		total for all years	species	feature	shapefile (.shp) GeoPackage (gpkg)	SpeciesPersistence_Annual_<TAXA>_<EAA>_<YYYY-YYYY>.shp (e.g. SpeciesPersistence_Annual_Plants_IBRA_1990-2024.shp)
		account for consecutive opening to closing year	species	data frame	comma-separated value (.csv)	SpeciesPersistence_Account_<TAXA>_<EAA>_<YYYY-YYYY>.csv (e.g. SpeciesPersistence_Account_Plants_IBRA_1990-2024.csv)
		account for consecutive opening to closing year	species	feature	shapefile (.shp) GeoPackage (gpkg)	SpeciesPersistence_Account_<TAXA>_<EAA>_<YYYY-YYYY>.shp (e.g. SpeciesPersistence_Account_Plants_IBRA_1990-2024.shp)
Proportion of species expected to persist	The proportion of originally occurring species in the reporting region expected to persist in the long term anywhere across their original range assuming no further change in ecosystem condition	total for all years	NA	data frame	comma-separated value (.csv)	ProportionPersistence_Annual_<TAXA>_<EAA>_<YYYY-YYYY>.csv (e.g. ProportionPersistence_Annual_Plants_IBRA_1990-2024.csv)
		total for all years	NA	feature	shapefile (.shp) GeoPackage (gpkg)	ProportionPersistence_Annual_<TAXA>_<EAA>_<YYYY-YYYY>.shp (e.g. ProportionPersistence_Annual_Plants_IBRA_1990-2024.shp)
		account for consecutive opening to closing year	NA	data frame	comma-separated value (.csv)	ProportionPersistence_Account_<TAXA>_<EAA>_<YYYY-YYYY>.csv (e.g. ProportionPersistence_Account_Plants_IBRA_1990-2024.csv)
		account for consecutive opening to closing year	NA	feature	shapefile (.shp) GeoPackage (gpkg)	ProportionPersistence_Account_<TAXA>_<EAA>_<YYYY-YYYY>.shp (e.g. ProportionPersistence_Account_Plants_IBRA_1990-2024.shp)

METRIC NAME	DESCRIPTION	SUMMARY TYPE	UNITS	DATA TYPE	DATA FORMAT	FILE NAME TEMPLATE (WITH EXAMPLE)
Ecosystem condition	Input data to the threatened species habitat (at 100 m resolution) and species persistence (at 250 m resolution)	annual	NA	raster	GeoTIFF (.tif)	EcosystemConditionInput_<YYYY>.tif (e.g. EcosystemConditionInput_2024.tif)

YYYY = year

YYYY-YYYY = year range of data (first year to last year)

TAXA = taxonomic group (‘Birds’; ‘Fungi’; ‘Plants’; ‘Reptiles’)

EAA = ecosystem accounting area (‘National’; ‘StateTerritory’; ‘SA2’; ‘SA2code’; ‘SA3’; ‘IBRA’; ‘IBRAsub’; ‘LGA’; ‘NRM’; ‘DrainageDivision’ ; ‘RiverRegion’ ; ‘ProtectedAreasAll’ ; ‘ProtectedAreasIUCN’; ‘Gridded50km’)

NA = not applicable

Table 4 Summary image file dictionary for biodiversity

METRIC NAME	DESCRIPTION	SUMMARY TYPE	UNITS	IMAGE TYPE	FILE NAME TEMPLATE (WITH EXAMPLE)
Threatened species habitat	Habitat provision for nationally listed threatened species	Annual amount	species.hectares	Map	ThreatenedSpeciesHabitat_<YYYY>.png (e.g. ThreatenedSpeciesHabitat_2024.png)
	Change in habitat provision for nationally listed threatened species over adjacent years	Change in annual amount	species.hectares	Map	ThreatenedSpeciesHabitatChange_<YYYY-YYYY>.png (e.g. ThreatenedSpeciesHabitatChange_2023-2024.png)
	National total habitat provision for nationally listed threatened species over time.	Time-series amount	species.hectares	Chart	ThreatenedSpeciesHabitat_National_1990-2024.png
Proportion of original threatened species habitat remaining	The proportion of original habitat provision for nationally listed threatened species	Annual proportion	NA	Map	ThreatenedSpeciesHabitatProportion_<YYYY>.png (e.g. ThreatenedSpeciesHabitatProportion_2024.png)
	Change in the proportion of original habitat provision for nationally listed threatened species over adjacent years	Change in annual proportion	NA	Map	ThreatenedSpeciesHabitatProportionChange_<YYYY-YYYY>.png (e.g. ThreatenedSpeciesHabitatProportionChange_2023-2024.png)
	National proportion of original habitat provision for nationally listed threatened species over time.	Time-series of proportion	NA	Chart	ThreatenedSpeciesHabitatProportion_National_1990-2024.png
Number of species expected to persist	National total number of originally occurring species in Australia expected to persist in the long term anywhere across their original range	Time-series amount	species	Chart	SpeciesPersistence_<TAXA>_National_1990-2024.png (e.g. SpeciesPersistence_Plants_National_1990-2024.png)
	Total number of originally occurring species in the ecosystem accounting area expected to persist in the long term anywhere across their original range	Annual amount	species	Map	SpeciesPersistence_<TAXA>_<EAA>_YYYY.png (e.g. SpeciesPersistence_Plants_SA2_2024.png)
	Change in the number of originally occurring species in the ecosystem accounting area expected to persist in the long term anywhere across their original range	Change in annual amount	species	Map	SpeciesPersistenceChange_<TAXA>_<EAA>_YYYY.png (e.g. SpeciesPersistenceChange_Plants_Gridded50km_2023-2024.png)
Proportion of species expected to persist	National proportion of originally occurring species in Australia expected to persist in the long term anywhere across their original range	Time-series of proportion	NA	Chart	ProportionPersistence_<TAXA>_National_1990-2024.png (e.g. ProportionPersistence_Plants_National_1990-2024.png)
	Proportion of originally occurring species in the ecosystem accounting area expected to persist	Annual proportion	NA	Map	ProportionPersistence_<TAXA>_<EAA>_YYYY.png (e.g. ProportionPersistence_Plants_Gridded50km_2024.png)

METRIC NAME	DESCRIPTION	SUMMARY TYPE	UNITS	IMAGE TYPE	FILE NAME TEMPLATE (WITH EXAMPLE)
	in the long term anywhere across their original range				
	Change in the proportion of originally occurring species in the ecosystem accounting area expected to persist in the long term anywhere across their original range	Change in annual proportion	NA	Map	ProportionPersistenceChange_<TAXA>_<EAA>_YYYY-YYYY.png (e.g. ProportionPersistenceChange_Plants_Gridded50km_2023-2024.png)

All images are provided in Portable Network Graphic (.png) format. NA = not applicable

YYYY = year

YYYY-YYYY = year range of data (first year to last year)

TAXA = taxonomic group (‘Birds’; ‘Fungi’; ‘Plants’; ‘Reptiles’)

EAA = ecosystem accounting area (‘National’; ‘StateTerritory’; ‘SA2’; ‘SA2code’; ‘SA3’; ‘IBRA’; ‘IBRAsub’; ‘LGA’; ‘NRM’; ‘DrainageDivision’ ; ‘RiverRegion’ ; ‘ProtectedAreasAll’ ; ‘ProtectedAreasIUCN’; ‘Gridded50km’)
NA = not applicable

2.5 Accuracy, assumptions and limitations

The data for biodiversity features provided here have a range of inherent assumptions and limitations. A full description of the accuracy, assumptions and limitations is available in the source documents referred to in the data processing methods in Section 2.2. Importantly, all of the data provided here reflect the potential habitat available for the different biodiversity features. A range of factors not considered in the models and analyses (e.g. metapopulation dynamics, diseases, time-lags in population dynamics and dispersal, dispersal barriers) may mean that those biodiversity features are not fully utilising some of the potential habitat being provided. This may also be influenced by our use of biological records from a relatively broad temporal span (1970−2025) to parameterise our models of expected biodiversity patterns as a function of environmental conditions. Note though that <10 % of biological records for birds and reptiles pre-date 1990.

All of the approaches applied rely on large amounts of ecological survey information, fine resolution static environmental layers, and dynamic layers derived from remote sensing. Where possible we have utilised systematic survey data over presence-only occurrence data, as they provide information on absences as well as presences of species. Errors in any of these data sources will result in subsequent errors in the predicted habitat for the biodiversity features considered.

Data on threatened species occurrences may be biased or influenced by redaction of some data by the data custodians. The threatened species habitat indicator is directly influenced by the identities of the species listed, hence biases and the level of effort in the species listing processes.

The spatial biodiversity models implemented will also have a range of inherent uncertainty, including those due to key drivers of biodiversity patterns or habitat quality being unknown and/or not included as predictor variables in the models. Where species-area relationships have been fitted based on species occurrence observations, the derived power-model scalar value (z) for each taxonomic group may be influenced by bias in the species occurrence observations. Future research could explore the way in which such bias may influence the inferred species-area relationships.

In interpreting and using the species persistence data, it is important to note that species persistence expressed as the number of originally occurring species requires the additional step of estimating the number of originally occurring species in each EAA. Given that these estimates of the number of native species originally occurring in each EAA add additional uncertainty and prediction error, species persistence expressed as a proportion of the originally occurring native species entails lower levels of uncertainty.

Certain aspects of the biodiversity data provided are also unable to be directly validated. In particular, the estimates of expected long-term species persistence for plants, reptiles, birds and fungi could only be directly validated centuries into the future, and only if ecosystem condition was maintained constant over time. However, the estimates for expected species persistence are based on spatial models of biodiversity patterns and ecosystem condition that have been validated, as well as foundational ecological theory (species-area relationships).

Given the importance of the ecosystem condition time series in the habitat-based assessment approach applied here, it is important to note that any spatial or temporal inaccuracies in the ecosystem condition data used (Valavi et al. 2025a; Williams et al. 2025) will directly influence the biodiversity time-series data that has been developed.

3 Data quality statement

3.1 Overview

Below we apply the Data Quality Framework developed by the ABS (ABS 2015) to evaluate seven dimensions of data quality for the biodiversity data developed and presented here for the purposes of SEEA EA. A full description of each dimension of data quality is provided in the Data Quality Framework (ABS 2015).

3.2 Institutional environment

The data products that comprise this collection were compiled by CSIRO, in collaboration and consultation with the ABS and DCCEEW. As Australia's national science agency, CSIRO has impartiality and objectivity, professional independence, national mandate, adequate resources, commitment to quality and statistical confidentiality necessary for developing these data to a high standard.

The data products were developed using the most credible, suitable, and up-to-date data sources that were available. These data sources were collected using:

- remotely sensed and on-ground measurements
- modelling techniques
- other published scientific datasets.

3.3 Relevance

This project updated the biodiversity data for ecosystem accounting in Australia from those presented in Mokany et al. (2024a). The data products cover the terrestrial areas of the Australian Continental EEZ at 100 m spatial resolution, and cover the period from 1990 to 2024, with reference to pre-1750.

Biodiversity can be quantified in many alternative ways. Here data are provided for two complimentary biodiversity indicators: threatened species habitat and species persistence. These indicators are particularly effective in integrating species-level biodiversity into the SEEA EA framework, with respect to both nationally listed threatened species and the persistence of all species within key taxonomic groups.

The data products are intended to support national ecosystem accounting and are aligned with SEEA EA recommendations on accounting for biodiversity (United Nations et al. 2024), particularly the alignment of the habitat-based approach applied here with section 13.3.4 of the SEEA EA. The data products provided here also align with published guidance on addressing spatial scale in deriving biodiversity accounts under SEEA Ecosystem Accounting (Larsen et al. 2021).

3.4 Timeliness

The data products are annual (calendar years) representing 1990 to 2024, with reference to pre-1750. The pre-1750 data serves as a reference for natural ecosystems prior to European colonisation. The year 1750 was selected as a commonly applied arbitrary pre-European year, and aligns with other data products such as the ecosystem extent data (Macfarlane et al. 2025) and the National Vegetation Information System (DCCEEW 2024a). The ecosystem condition data used as an input for each year incorporates the preceding two years of remotely sensed data, along with the reporting year (e.g. the data layer for 2024 includes remote sensing input from the years 2022, 2023 and 2024) (Valavi et al. 2025a; Williams et al. 2025). This means that interpreting change between adjacent years should consider the potential factors influencing that change across the preceding two years.

3.5 Accuracy

Factors influencing the accuracy of the data time series developed here are documented in Section 2.5 . As noted there, the biodiversity data that have been developed are impacted by the accuracy, assumptions and limitations associated with each of the source datasets used, as documented in the metadata of those source datasets. The accuracy of those source datasets are fully documented in citations provided throughout Section 2.2.

3.6 Coherence

The methods typologies used are consistent with the SEEA EA framework (United Nations et al. 2024) and thus coherent for the purposes of international reporting and comparison. The datasets used as inputs are national scale, to maximise consistency across state and territory boundaries. Both biodiversity indicators provided here use as an input the same time series of ecosystem condition (1990-2024), which itself is generated in a consistent manner over that period, using remotely sensed data from the Landsat product, curated by Geoscience Australia (Valavi et al. 2025a; Williams et al. 2025). The input biodiversity data applied are sourced from national curated collections of biodiversity observations (Atlas of Living Australia 2025; Bissett et al. 2016; Mokany et al. 2022a). The threatened species habitat indicator applies national data on the species listed as threatened and their estimated distributions (DCCEEW 2025a). The species persistence indicator is recognised as a Component Indicator (as the 'Biodiversity Habitat Index') in the Convention on Biological Diversity Kunming-Montreal Global Biodiversity Framework (CBD 2022; UNEP-WCMC 2023). The Ecosystem Accounting Areas used for deriving the species persistence indicator data are established and curated national regionalisations for Australia, including legally recognised jurisdictions (Section 2.2.2).

Key datasets used as inputs are developed and maintained by Commonwealth Government agencies to ensure coherence at a national scale. Ongoing development of all datasets may result in small changes between versions of the time series within Australia.

3.7 Interpretability

The data products are intended to support the creation of national ecosystem accounts, and are based on the SEEA EA framework (United Nations et al. 2024). They are not intended as a substitute for more detailed state or regional scale data, if such data exists, and users should consider whether the data products are suitable for their purpose.

This technical report details the methods used to create the data products and includes reference to relevant scientific literature and/or related links that can provide extensive additional background information to support interpretation, in addition to a glossary of scientific and technical terms. Both biodiversity indicators provided have meaningful units of measure that are directly understandable and interpretable. The key source data influencing spatiotemporal changes, the ecosystem condition data (developed in HCAS v3.3 and resampled here), are provided with the data collection to aid interpretability as to when and where change in the biodiversity indicators occurs.

To help users interpret this data product, datasets published on the CSIRO Data Access Portal include metadata (e.g. collection description, bounding box, publication date, authorship, licencing). This data product includes graphical summaries of the data, including map images and charts, to enhance the interpretability of the data. The data is structured and labelled in a way to ensure each file is interpreted correctly.

3.8 Accessibility

The data products have been published under a Creative Commons Attribution 4.0 International (CC BY 4.0) licence and are freely accessible from CSIRO's Data Access Portal. The collections published as part of this project are provided in formats that can be accessed by both open source and proprietary software, supporting accessibility and reusability. Complementary metadata (such as provided in CSIRO's Data Access Portal fields) provide semantic definitions of the data content, supporting both accessibility and interoperability between systems. Contact details for requesting more information are also given on the CSIRO Data Access Portal.

Appendix A Summary of evidence relating threatened species habitat quality to ecosystem condition

A.1 Introduction

Degradation and loss of habitat are among the greatest threats facing biodiversity, affecting common and threatened species alike. Over one million species are at risk of extinction (IPBES, 2019), with more than half estimated to have insufficient habitat for long-term survival (Leclère et al. 2020). Within Australia, threatened species are estimated to have lost half of their historical habitat (Giljohann et al. 2025), and nearly 70 % of species are directly impacted by habitat degradation and loss (Ward et al. 2021).

Multiple studies have demonstrated the importance of habitat quality for individual survival, reproductive success and body condition (Bennett et al. 2012; Hale et al. 2019; Thomas et al. 2021; van Doorn et al. 2015). However, as the measures used to assess habitat quality are generally species or study specific, this limits the ability to draw comparisons between studies, let alone infer whether threatened species respond to changes in broader ecosystem condition in similar ways. Understanding whether species, or taxonomic groups, respond similarly to habitat quality would provide a basis for generalising these relationships to other unstudied species.

Ecological theory suggests that there exists threshold limits for habitat properties whereby if a threshold is crossed, the population will precipitously and suddenly decline (Hanski and Ovaskainen 2000; With and King 1999). This theory was applied in the design of the threatened species habitat indicator, where it was assumed that sites below a threshold level of ecosystem condition (a surrogate for habitat quality) provide no habitat for threatened species, and above this value the benefit of the habitat increases linearly with the ecosystem condition score (Giljohann et al. 2025).

Previous derivation of the threatened species habitat indicator used an ecosystem condition threshold (Figure 1) (Giljohann et al. 2025; Mokany et al. 2024b). This study aims to provide an empirical basis to inform the design of the threatened species habitat function and tests the existence of an ecosystem condition threshold below which the habitat does not support threatened species. To do this we first undertook a review of the published evidence relating threatened species population size to ecosystem condition. We then carried out a quantitative analysis using threatened species abundance data sourced from the Australian Threatened Species Index and ecosystem condition data sourced from CSIRO's Habitat Condition Assessment System[3].

[3] https://research.csiro.au/biodiversity-knowledge/projects/hcas/

A.2 Methods

Literature review

We conducted structured literature searches between 15 April and 10 March, 2025 using Web of Science. Six literature searches were undertaken, using different combinations of keywords (see Table A.1). Based on article title and abstract, our initial vetting retained articles that were: (i) terrestrially based (not marine), and; (ii) likely to contain data on population size or growth rate or individual condition or fitness, as well as some qualification of habitat quality. Articles that were considered likely to contain relevant data became "articles after initial vetting" which were read entirely and searched for relevant data.

Table A.1 Keywords, number of articles returned, number of articles after vetting for 6 literature searches for published data relating threatened species populations to habitat quality

Keywords	Articles returned	After initial vetting	Articles with relevant data
"habitat", "quality","population", "Australia"	292	31	1
"long", "term", "monitoring", "population", "habitat quality", "threatened"	82	7	0
"long", "term", "monitoring", "Australia", "population", "quality", NOT: "cancer", "trauma", "psychol*", "medicin*"	217	4	0
"threatened", "species", "habitat", "quality", "Australia"	278	13	0
"threatened species", "habitat quality", "Australia"	86	6	0
"effect", "habitat quality", "population", "threatened"	169	9	1

Overview: threatened species population data related to ecosystem condition mapping

To relate the threatened species abundance data to the ecosystem condition data, we developed a statistical model of expected abundance as a function ecosystem condition at the monitoring site in the survey year.

Threatened Species data

Threatened species monitoring data were sourced from the Australian Threatened Species Index (TSX) on March 24, 2025. The TSX compiles a continental-scale database of thousands of threatened species monitoring projects, contributed by over 30 organisations, which it uses to derive an index of change in species relative abundances over time (TSX 2025). The database consists of site-based monitoring records for amphibians, birds, mammals and plants spanning 1950 to present.

The publicly available data is provided as a timeseries of species abundances for each monitoring site, with site location generalised to the IBRA region in which it is situated. Survey methods, reporting units and monitoring frequencies vary considerably between species and between sites but must be consistent within a timeseries (species x site).

The data were filtered to retain timeseries that: (i) reported abundance as counts (i.e. excluding proportions, percentages, reporting rates); (ii) were based on survey methods likely to be less affected by detectability (e.g. trapping, plot survey); (iii) surveyed a defined spatial area that

related to the spatial extent of the ecosystem condition data (e.g. excluding aerial surveys, counts aggregated across multiple sites, range wide surveys), and; (iv) overlapped the HCAS ecosystem condition data both spatially (e.g. excluding marine taxa) and temporally (excluding years prior to 1990 and post 2022). Sites tagged as actively managed (e.g. nest boxes added) were also omitted as these sites may exhibit different dynamics to unmanaged sites. Finally, we omitted counts that were outliers within their taxonomic group, for plants this was counts >1000 individuals and for and amphibians this was counts >100 individuals. The filtered data covered eight amphibian species, three bird species, nine mammal species and seven plant species (Table A.2).

Table A.2 Species included in the filtered TSX data and the number of records (species x site x year).

Species	Amphibians	Birds	Mammals	Plants
Litoria aurea	78			
Litoria olongburensis	63			
Litoria raniformis	64			
Litoria watsoni	54			
Philoria frosti	95			
Pseudophryne corroboree	80			
Pseudophryne pengilleyi	80			
Spicospina flammocaerulea	89			
Climacteris picumnus victoriae		71		
Dasyornis brachypterus monoides		77		
Pezoporus wallicus wallicus		143		
Chalinolobus picatus			68	
Conilurus penicillatus			83	
Dasyurus hallucatus			238	
Isoodon obesulus obesulus			91	
Mesembriomys gouldii melvillensis			78	
Petaurus australis			63	
Phascogale tapoatafa			194	
Phascolarctos cinereus			63	
Sarcophilus harrisii			70	
Acacia menzelii				51
Acacia rhetinocarpa				79
Caladenia colorata				75
Grevillea calliantha				92
Olearia pannosa subsp pannosa				53
Prostanthera eurybioides				52
Thelymitra epipactoides				81

Ecosystem condition data

As data on the ecosystem condition of monitoring sites is not collated in the TSX database we used data from the Habitat Condition Assessment System (HCAS) to provide a consistent continental-wide time series of ecosystem condition (Harwood et al. 2016; Valavi et al. 2025b). HCAS integrates biological and physical ecosystem data with remote sensing data to generate ecosystem condition scores that vary from 0.0 (ecosystem completely removed or transformed) to 1.0 (ecosystem with the highest ecological integrity). We used HCAS v3.1, which has a 90 m spatial resolution and was calculated as three-year rolling averages from 1988 to 2023 (Valavi et al. 2025b). More recently published versions of HCAS were not available at the time of analysis.

HCAS data were joined to the TSX time series by the TSX data manager using the actual site coordinates (not included in the publicly available data). Each survey year in the TSX timeseries (species x site x year) was associated with the correct year in the HCAS time series. Data were extracted as the ecosystem condition value of the grid cell underlying the site coordinates, and the average ecosystem condition (of non-NA grid cells) across a neighbourhood of 3 x 3 grid cells

centred on the site coordinates (i.e. 270 x 270 m). A neighbourhood measure was selected to account for the fact that the size and orientation of the survey areas varies considerably and often exceeds the 90 m grid cell of the HCAS resolution (pers. comm. TSX team), plus there is inherent spatial uncertainty in both the TSX location data and the remotely-sensed data used as input to HCAS.

The TSX time series was subsequently filtered to: (i) omit records (species x site x year) where there was large dissimilarity between the site condition and neighbourhood condition (sites with less than median similarity were omitted), and; (ii) omit species x survey type (TSX field: SearchTypeDesc) that sample only a narrow range of ecosystem condition values (<0.5) or have <10 sites. These filters removed sites or groups where the condition values were unrepresentative (i.e. statistical outliers) or didn't provide sufficient information across a gradient of ecosystem condition values to fit informative relationships.

The final dataset consisted of 2,590 records (species x site x year), representing eight amphibians, three birds, nine mammals and seven plants (Table A.2, Figure A.1). The abundance data were rounded to integers.

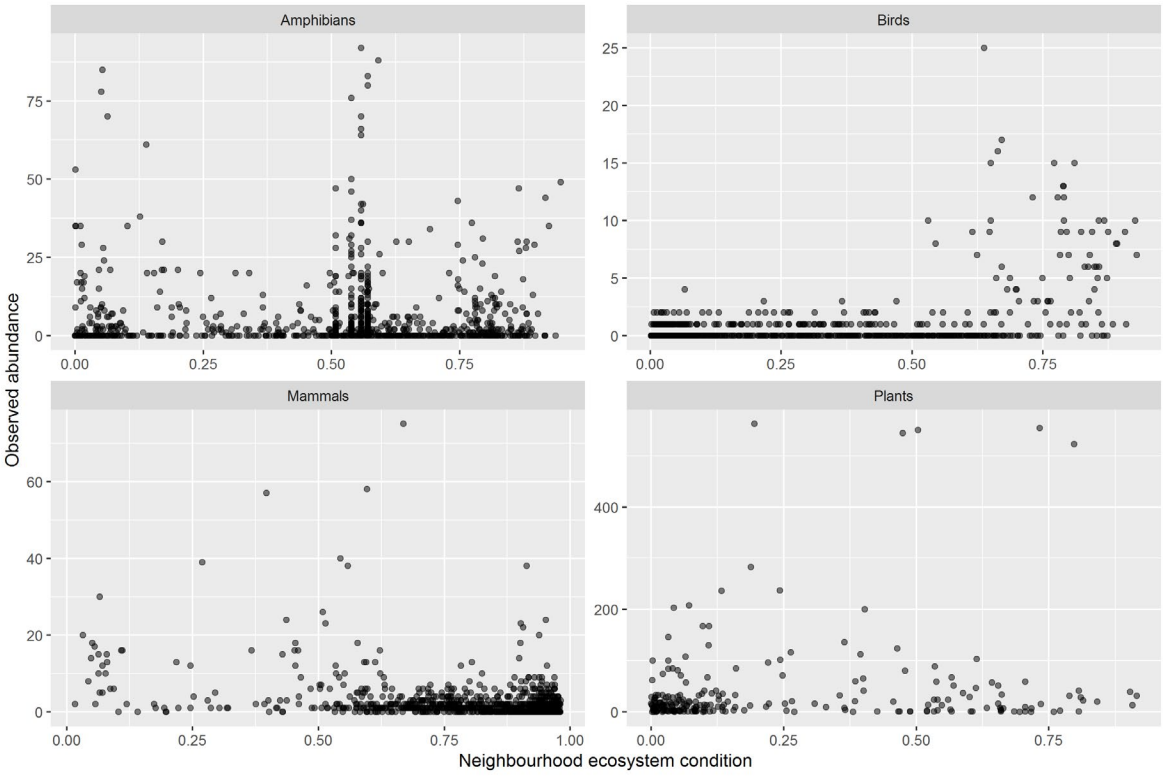

Figure A.1 Abundances for each record (species x site x year) in the filtered TSX data by neighbourhood ecosystem condition

Modelling

To analyse the TSX and HCAS data we used a hurdle model to account for the large number of surveys with zero abundances. Hurdle models treat zero-count and non-zero outcomes as two completely separate categories. The probability of a zero-count was modelled using a binomial distribution with a logit link as a function of the taxonomic group (Ts) of species s and the average neighbourhood ecosystem condition ($ECit$) at site i in year t. $ECit$ was fitted as a polynomial of degree 3 to allow a non-linear relationship between ecosystem condition and abundance, and an

interaction between Ts and $ECit$ allows the response to ecosystem condition to vary between taxonomic groups.

The non-zero outcomes were modelled using a truncated negative binomial distribution with a log link, a random intercept and slope to account for extra variation among species, random effects for site nested in IBRA region and for year, and additional dispersion by taxonomic group. The model defined the expected abundance $Asit$ of species s at site i in year t as a function of taxonomic group (Ts) and ecosystem condition ($ECit$). The explanatory variables $ECit$, and the interaction Ts:$ECit$ were modelled in the same way as in the zero-count model. The random effects were drawn from a normal distribution with mean zero and a standard deviation that was estimated from the data.

Models were run in R version 4.4.1 (R Core Team 2025). The hurdle model was fitted using the glmmTMB package (McGillycuddy et al. 2025). Model diagnostics were calculated using package DHARMa (Hartig 2024) and plots were created using the package ggplot2 (Wickham 2016).

In model fitting we also investigated alternate model specifications, including a truncated Poisson distribution, zero-inflated Poisson or zero-inflated negative binomial distribution, a model of relative abundance using a zero-inflated ordinal beta distribution, and a logistic model of species presence and absence. The alternate model specifications were rejected in favour of the hurdle model with a truncated negative binomial distribution on the basis of model diagnostics. These results are not presented here.

A.3 Results

Literature review

Very few studies were found that directly measured threatened species population size or similar measure in relation to ecosystem condition or habitat quality. There was broad consensus in the literature that habitat quality influences species survival, and therefore the success of conservation efforts depend on conserving areas of high habitat quality (Hamer and Parris 2011; Lawton et al. 2021; Meggs et al. 2004; Sheean et al. 2012; van Doorn et al. 2015; Verdon et al. 2025; With and King 1999; Wolf et al. 1996).

Quantitative analysis of TSX and HCAS data

The relationship between threatened species abundance and ecosystem condition was predicted to differ markedly between species and between taxonomic groups. Model diagnostics supported the hurdle model as providing a good fit to the data (Figure A.2, Figure A.3). Predicted abundances were slightly higher than those observed for low abundances, and slightly lower than those observed for higher abundances (Figure A.2).

The zero-count model predicted amphibian and bird species to have a high probability of being absent regardless of ecosystem condition (Figure A.4), although bird species were more likely to be predicted absent at lower conditions. By contrast mammal and plant species had a low probability of being predicted absent, yet for mammal species the probability of absence increased to almost 50 % as ecosystem condition approached 1 (Figure A.4).

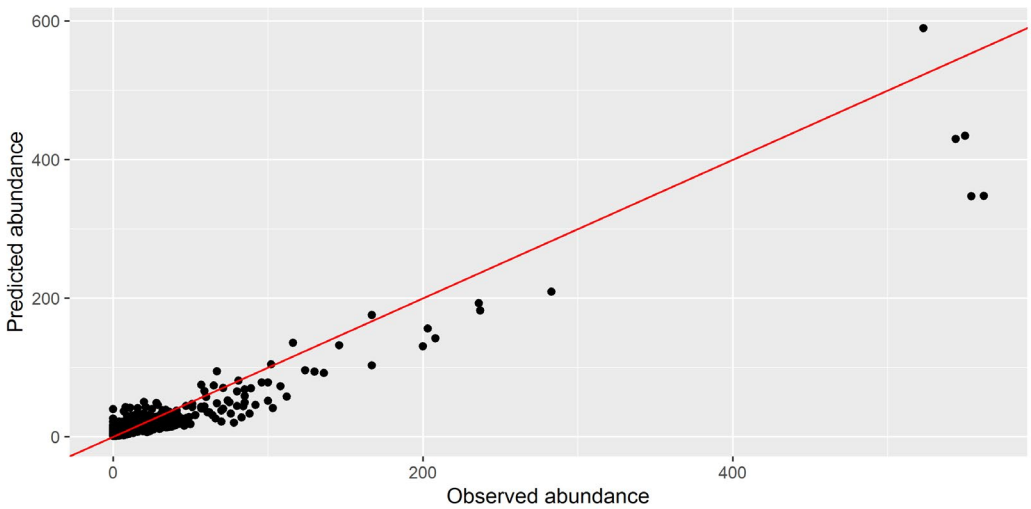

Figure A.2 Diagnostic plots for the hurdle model. Predicted non-zero abundance plotted against the observed abundance data

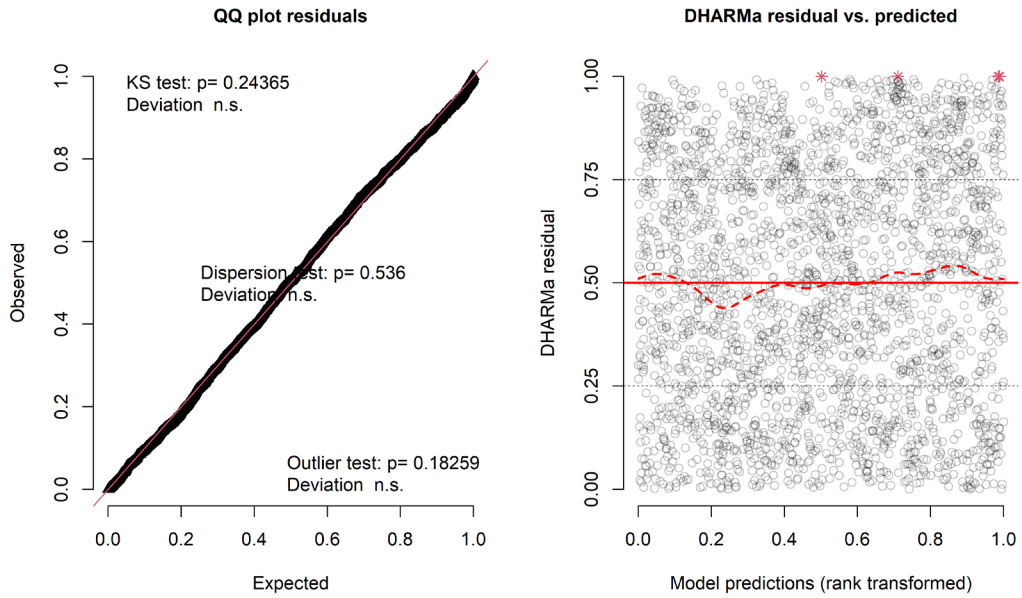

Figure A.3 Diagnostic plots for the hurdle model. Residual plots show the fitted model does not violate the assumptions of the statistical model

The population-level predictions of abundance as a function of ecosystem condition varied from predominantly null for amphibians, to negative for mammals, and positive for both birds and plants (Figure A.5). However, the influence of ecosystem condition on abundance was only statistically significant for plants and birds (Box 1), reflecting the high level of within taxonomic group variation (Figure A.6). The full table of model coefficients is provided in Box 1.

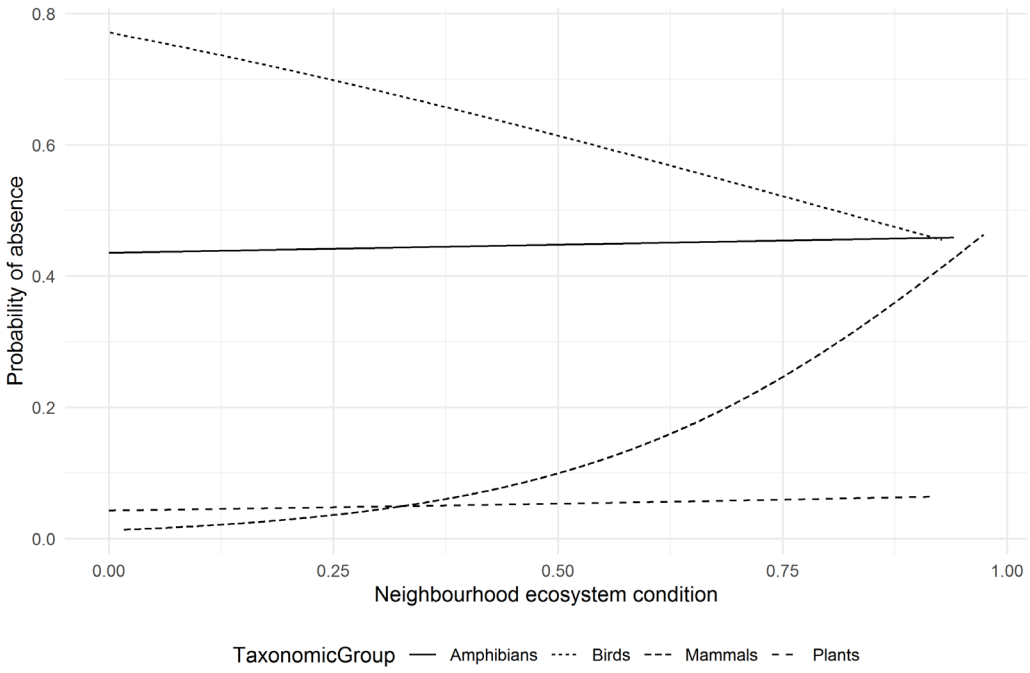

Figure A.4 Predicted probability of a zero-count as a function of taxonomic group and ecosystem condition

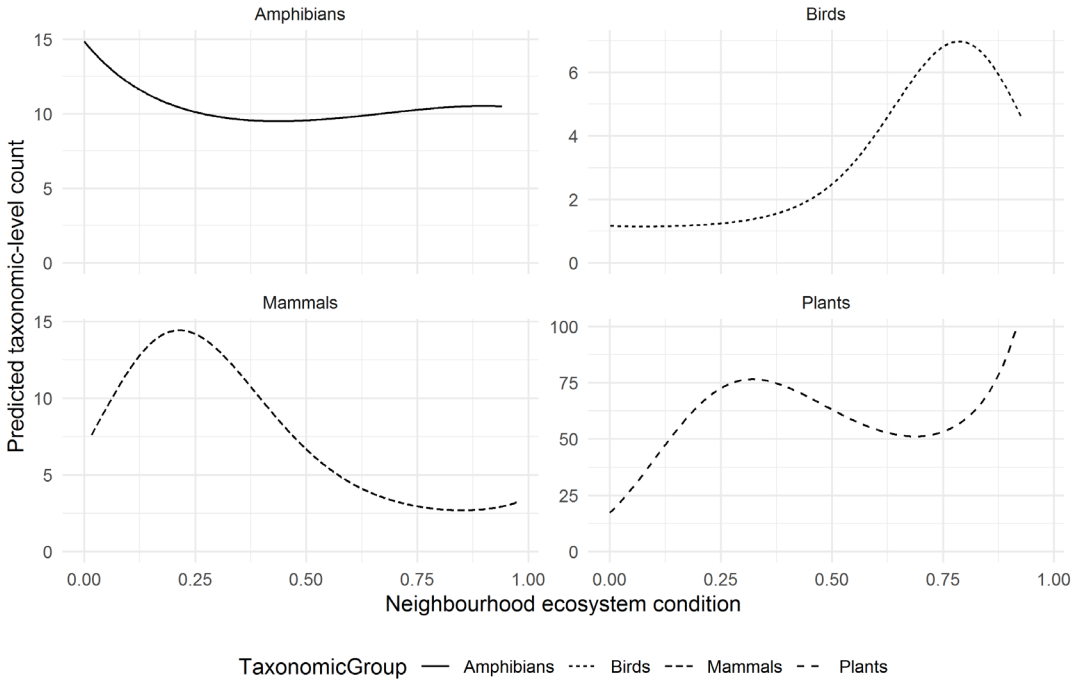

Figure A.5 Predicted mean population-level abundance as a function of taxonomic group and ecosystem condition. All random effects have been set to zero. Note the change in range of abundances (y-axis) between taxonomic groups.

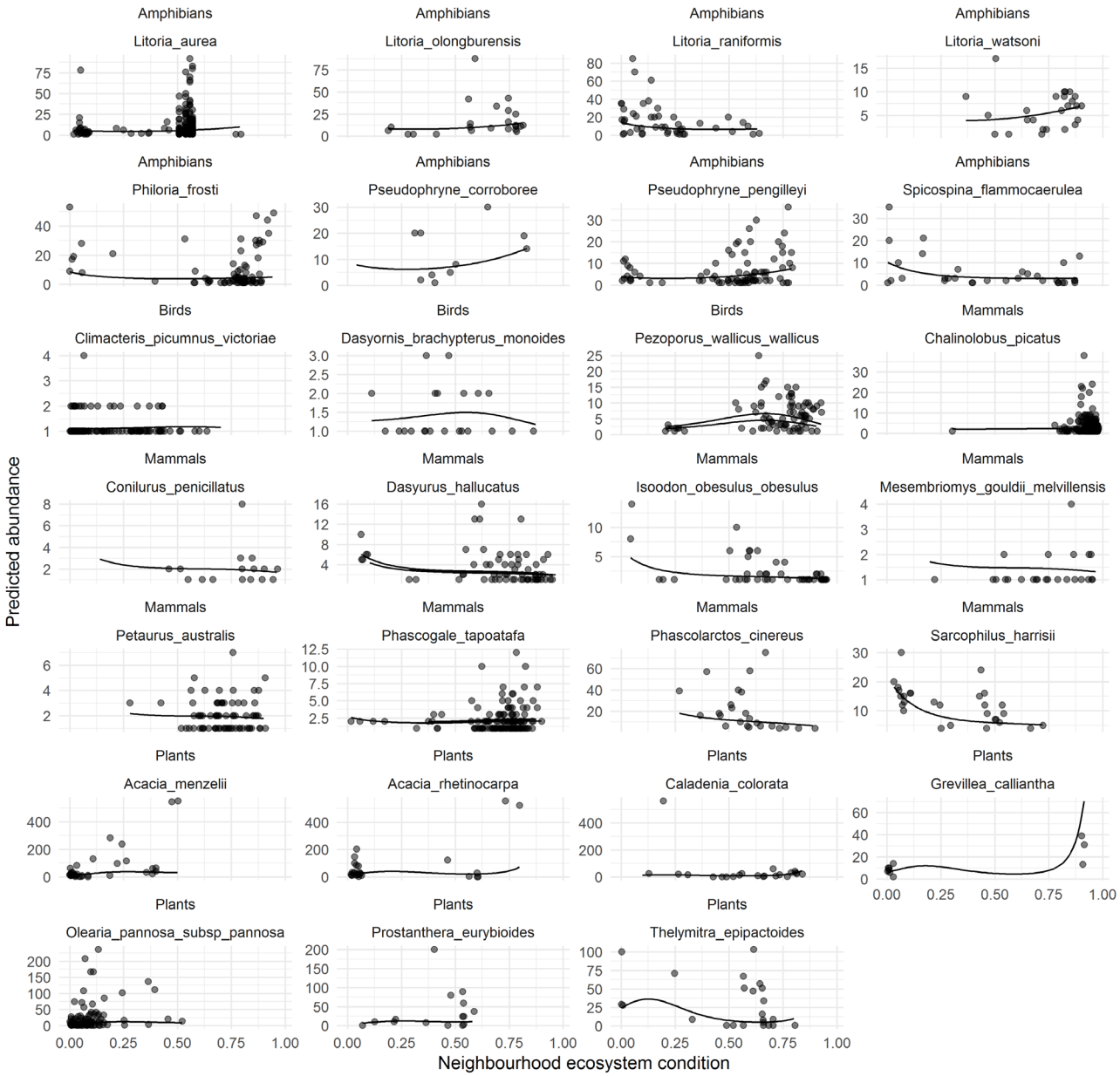

Figure A.6 Predicted mean abundance for each species as a function of ecosystem condition. Note the change in range of abundances (y-axis) between the panels.

A.4 Discussion

Our analysis of the TSX monitoring data and HCAS-derived ecosystem condition did not support the existence of a threshold in ecosystem condition below which habitat is too marginal for threatened species to occur (Figure 1). For two of the four taxonomic groups, the predicted association between abundance and ecosystem condition upheld the assumption (underpinning the threatened species indicator) that the value of habitat increases with increasing ecosystem condition. The model results were inconclusive for the other two taxonomic groups.

The predicted association between abundance and neighbourhood ecosystem condition exhibited a variety of responses, both within and between taxonomic groups. At the taxonomic-level, the model predicted that bird species were more likely to be absent at sites of lower ecosystem condition, and that, on average, their abundance is expected to be higher at sites where the

ecosystem condition is higher – although the overall change in abundance was small. Note that only three bird species were included in the modelling. Plant species were predicted to have the strongest relationship with ecosystem condition, in terms of both statistical significance and effect on abundance; with expected abundance increasing as ecosystem condition increased. The probability of plant absence was predicted to be low across the full ecosystem condition range.

For amphibian species, ecosystem condition was not predicted to effect either their absence or abundance. This taxonomic-level result is likely due to the contrasting responses exhibited by individual species, for example *Litoria raniformis* and *Spicospina flammocaerulea* had highest abundance at lower condition sites, *Litoria watsoni* had highest abundance at high condition sites, and other species displayed intermediate responses. The response of mammal species to ecosystem condition diverged from the other taxa. The model predicted mammal species were more likely to be absent at sites where the ecosystem condition was higher, and that on average their abundance was expected to be highest at sites of lower ecosystem condition – although noting that the effect of ecosystem condition on non-zero abundance was not statistically significant. This unintuitive outcome is potentially an artefact of the data used in the modelling (discussed below), rather than an actual biological response.

In interpreting the results of this study there are some key factors to take into consideration. First, we used available data. We sourced data from the TSX database, a key national resource of threatened species population data, which is curated to provide information for assessing change in species' populations over time. Each timeseries needs to use a consistent monitoring method and unit of population measurement, but these can, and do, vary significantly between species and between monitoring programs for the same species. This presented some challenges for the analysis, and we took multiple steps to account for known sources of variation. This included filtering, using a neighbourhood measure of ecosystem condition to account for variation in survey size and orientation, ensuring sub-groups within the data (species x survey type) encompassed a minimum span of the condition range; using a method capable of modelling excess zeros and over-dispersed counts; and interrogating model results by comparing to an equivalent model of relative abundance. However, without additional information we were not able to disentangle whether the large number of absences, as recorded for many of the mammal species at sites with high ecosystem condition, are true absences or result from variation in sampling effort and detectability (McCarthy et al. 2013) (i.e. it may be more difficult to detect mammal species in ecosystems with high plant biomass across structural layers). The relationship for mammals is also potentially influenced by biased sampling, if low condition sites are only surveyed when there is a known high likelihood of observing the threatened species.

Monitoring of threatened species will, by nature, target sites where the species is known to occur. As is evident in both the raw abundance data and model predictions, threatened species occur across the full range of ecosystem condition. It is important to note that from the perspective of this analysis, sites that are strongholds of the species but are not otherwise representative of high-quality habitat can produce misleading results. For example, combining data from last known populations holding on in cities or other highly modified areas (Soanes and Lentini 2019), with surveys for additional populations in previously unrecorded but high-quality habitat will confound attempts to decipher a logical relationship. Systematic surveys across the condition gradient would be needed for a rigorous evaluation.

The gold standard data for this analysis would additionally include on-ground measurements of ecosystem condition collected at the same time as population survey. However, ecosystem condition (or habitat quality) data is not typically collected in the monitoring of threatened species populations and is not collated in the TSX database (TSX 2025). In lieu of on-ground data, we used the HCAS ecosystem condition data as it provides consistent national-scale mapping across a time series and at fine spatial resolution (90 m) (Valavi et al. 2025b). To obtain the most relevant data for the analysis we were assisted by the TSX team who extracted the HCAS time-series data to the site coordinates. HCAS is based on vegetation indices derived from remote sensing, and despite it not incorporating other habitat modifying factors that have a large impact on threatened species (e.g. weeds, rabbits, feral cats) (Legge et al. 2023; Woinarski et al. 2015) it was able to predict changes in abundance for both bird and plant species. As HCAS is primarily a terrestrial measurement of ecosystem condition it only partially reflects amphibian habitat, which may in part explain the predicted null response of amphibian species to ecosystem condition. An important and related point, is that most of Australia's threatened species are typically affected by multiple threats (Kearney et al. 2023; Legge et al. 2023). Threats such as disease and genetic bottlenecks impact species abundances irrespective of habitat quality.

This study aimed to provide an empirical basis to inform the design of the function relating threatened species habitat to ecosystem condition. The data did not support the existence of a threshold in ecosystem condition below which threatened species do not occur (Figure 1). Extinction debts and other demographic lags can result in populations hanging on in habitat that cannot support the species in the long term (Kuussaari et al. 2009). A rigorous test for a biologically meaningful threshold would require a measure of population persistence (Akcakaya and Sjögren-Gulve 2000; McCarthy et al. 2014). We did, however, find support for a positive relationship between species abundance and ecosystem condition for two of four taxonomic groups, with the other two taxonomic groups being inconclusive. On this basis of these results, it is recommended that the threatened species indicator not apply a threshold in ecosystem condition and instead implement a simple linear 1:1 relationship that increases from ecosystem condition of 0 to 1 (Figure 1).

Box 1 Summary of models fitted: TSX data ~ HCAS condition

Family: truncated_nbinom2 (log)
Formula: Count ~ TaxonomicGroup * poly(ConditionAvg9Cell, 3) + (1 + ConditionAvg9Cell | Binomial) + (1 | SearchTypeDesc) + (1 | Region/SiteID) + (1 | Year)
Zero inflation: ~TaxonomicGroup * ConditionAvg9Cell
Dispersion: ~TaxonomicGroup
Data: d_prep1

AIC	BIC	logLik	-2*log(L)	df.resid
11433.7	11642.4	-5681.8	11363.7	2837

Random effects:
Conditional model:

Groups	Name	Variance	Std.Dev.	Corr
Binomial	(Intercept)	1.185830	1.08896	
	ConditionAvg9Cell	2.645805	1.62659	-0.59
SearchTypeDesc	(Intercept)	0.086250	0.29368	
SiteID:Region	(Intercept)	0.984365	0.99215	
Region	(Intercept)	0.080495	0.28372	
Year	(Intercept)	0.007314	0.08552	

Number of obs: 2872, groups: Binomial, 27; SearchTypeDesc, 18; SiteID:Region, 881; Region, 57; Year, 32

Conditional model:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	1.7423	0.4210	4.139	3.49e-05 ***
TaxonomicGroupBirds	-2.1101	0.7857	-2.685	0.007243 **
TaxonomicGroupMammals	-0.9754	0.5600	-1.742	0.081551 .
TaxonomicGroupPlants	1.5393	0.6675	2.306	0.021099 *
poly(ConditionAvg9Cell, 3)1	5.0920	13.7636	0.370	0.711411
poly(ConditionAvg9Cell, 3)2	11.7670	7.9835	1.474	0.140505
poly(ConditionAvg9Cell, 3)3	-2.5742	5.5456	-0.464	0.642509
TaxonomicGroupBirds:poly(ConditionAvg9Cell, 3)1	-5.8392	26.1242	-0.224	0.823133
TaxonomicGroupMammals:poly(ConditionAvg9Cell, 3)1	-29.2796	19.6179	-1.492	0.135569
TaxonomicGroupPlants:poly(ConditionAvg9Cell, 3)1	45.8373	22.3016	2.055	0.039847 *
TaxonomicGroupBirds:poly(ConditionAvg9Cell, 3)2	-34.5103	14.9670	-2.306	0.021124 *
TaxonomicGroupMammals:poly(ConditionAvg9Cell, 3)2	-5.4932	11.1985	-0.491	0.623755
TaxonomicGroupPlants:poly(ConditionAvg9Cell, 3)2	32.4207	16.0642	2.018	0.043571 *
TaxonomicGroupBirds:poly(ConditionAvg9Cell, 3)3	-5.2200	9.4752	-0.551	0.581691
TaxonomicGroupMammals:poly(ConditionAvg9Cell, 3)3	-2.5986	7.7979	-0.333	0.738954
TaxonomicGroupPlants:poly(ConditionAvg9Cell, 3)3	29.5978	8.8850	3.331	0.000865 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-0.2592	0.1589	-1.631	0.103
TaxonomicGroupBirds	1.4766	0.2019	7.312	2.63e-13 ***
TaxonomicGroupMammals	-4.1158	0.5211	-7.898	2.84e-15 ***
TaxonomicGroupPlants	-2.8497	0.4624	-6.163	7.14e-10 ***
ConditionAvg9Cell	0.1007	0.2785	0.361	0.718
TaxonomicGroupBirds:ConditionAvg9Cell	-1.6068	0.4045	-3.972	7.11e-05 ***
TaxonomicGroupMammals:ConditionAvg9Cell	4.2414	0.6365	6.664	2.67e-11 ***
TaxonomicGroupPlants:ConditionAvg9Cell	0.3589	1.1579	0.310	0.757

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Dispersion model:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	0.3342	0.1368	2.444	0.01454 *
TaxonomicGroupBirds	3.5677	2.0972	1.701	0.08892 .
TaxonomicGroupMammals	0.7737	0.2494	3.103	0.00192 **
TaxonomicGroupPlants	1.1547	0.2616	4.415	1.01e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Glossary

TERM	DEFINITION
Basic spatial unit (BSU)	the smallest unit of aggregation that is used consistently across accounts and against which stocks and flows are recorded
Biodiversity	the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species, and of ecosystems (CBD 1992)
Community-level biodiversity	consideration of biodiversity for an assemblage of species within a taxonomic group at a location
Ecosystem	a dynamic complex of plant, animal and micro-organism communities and their non-living environment interacting as a functional unit (CBD 1992)
Ecosystem accounting area (EAA)	the geographical territory for which an ecosystem account is compiled (United Nations et al. 2024)
Ecosystem asset	a contiguous space of a specific ecosystem type characterized by a distinct set of biotic and abiotic components and their interactions (United Nations et al. 2024)
Ecosystem condition	the quality of an ecosystem measured in terms of its abiotic and biotic characteristics (United Nations et al. 2024)
Ecosystem extent	the area of an ecosystem asset in terms of spatial area (United Nations et al. 2024)
Ecosystem functional groups (EFG)	third level of the International Union for Conservation of Nature (IUCN) Global Ecosystem Typology (GET) classification. They are functionally distinctive groups of ecosystems within a biome and are defined in a manner consistent with the CBD definition of ecosystems (United Nations et al. 2024).
Effective habitat area	The estimated total area of habitat within a region available for biodiversity, considering the ecosystem condition of each location (grid cell) within that region. Calculated as the area of the region multiplied by the average ecosystem condition of all grid cells within that region.
Exchange values	A measure of the equivalent monetary value of an ecosystem service that does not have an explicit market price. The exchange value is an estimate of what the market price would be if a market existed for the service.
Habitat Condition Assessment System	a method to remotely assess and map the generalised condition of natural habitat for terrestrial native biodiversity at a location against a reference condition derived from the dynamics of the most intact examples of native vegetation / ecosystems across contemporary Australia (Williams et al. 2021). Output is equivalent to an overall index of ecosystem condition in the SEEA EA framework
IUCN	International Union for Conservation of Nature
Potential extent of occurrence	the area contained within the shortest continuous imaginary boundary that can be drawn to encompass all the current known localities, as well as inferred occurrence and projected original occurrence of a species (Brooks et al. 2019)
SEEA EA	System of Environmental Economic Accounting - Ecosystem Accounting
Species-level biodiversity	consideration of biodiversity for each individual species separately
Species.hectares	unit for threatened species effective habitat and defined as the number of threatened species multiplied by the effective habitat area
Species persistence	the ongoing maintenance of a species as viable populations over the long term
Species richness	the number of species occurring in a location, typically considered within a specific taxonomic group
Thematic account	Standalone accounts, or sets of accounts, that organise data around specific policy-relevant themes, such as accounting for biodiversity, carbon and oceans, or protected areas, wetlands, forests and urban areas.
Trend	qualitative or quantitative assessment of the general direction of change (e.g. increasing, decreasing, steady) that is developing based on a time-series of measurements

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